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MIL-F-16377/56(SHIPS)
31 August 1971
SUPERSEDING
MS17268(SHIPS)
15 May 1963
(See page 4)

MILITARY SPECIFICATION SHEET

FIXTURE, LIGHT, INCANDESCENT, 100-WATT, 120-VOLT,

MACHINE TOOL, PORTABLE, FLOATING ADJUSTMENT

SYMBOL 266.2

The complete requirements for procuring the light described herein shall consist of this document and the latest issue of MIL-F-16377(SHIPS).

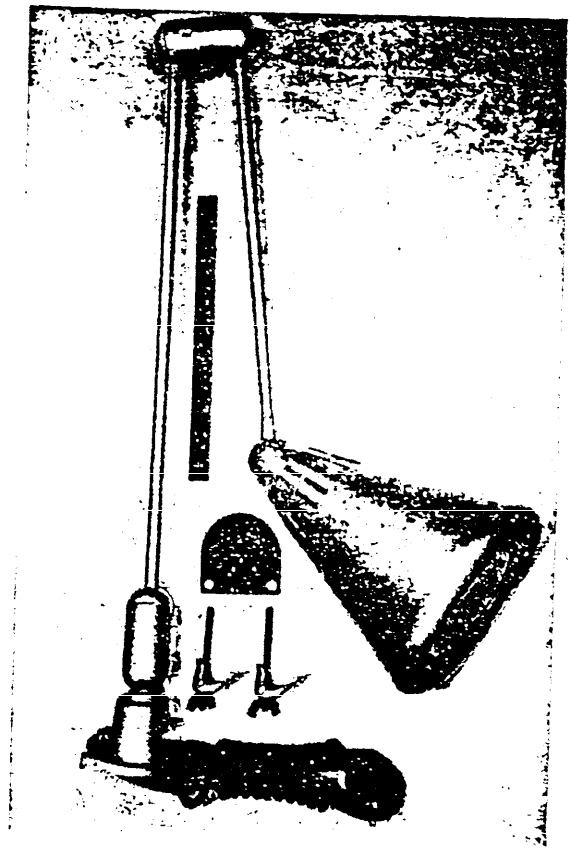
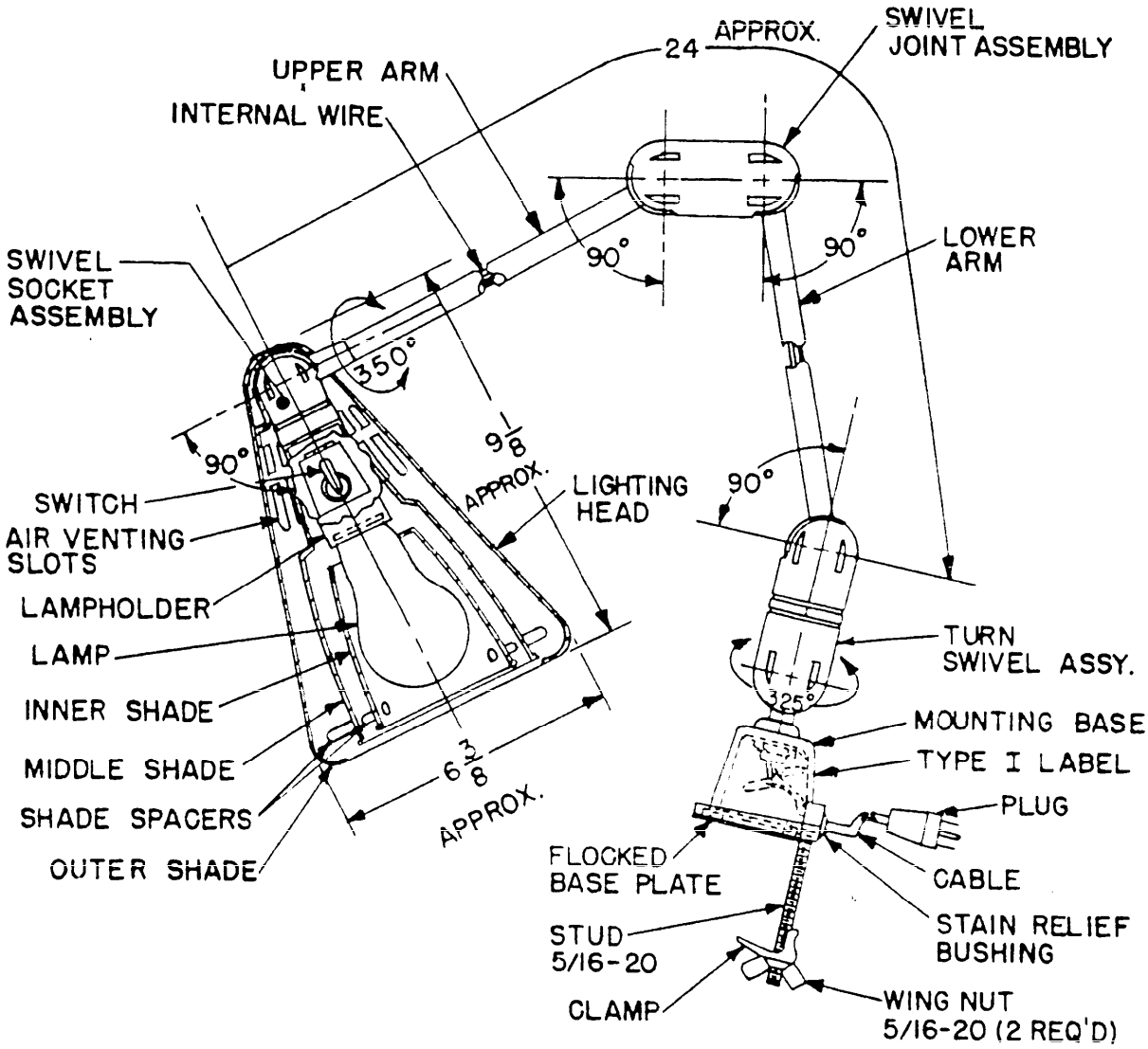


Figure 1 - Machine tool light, symbol 266.2.

All dimensions in inches.
For tolerances other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6230

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SH 10271

Figure 2 - Section views, machine tool light, symbol 266.2

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REQUIREMENTS:

Type:	II
Class:	1
Symbol 266.2:	Light, incandescent, 100-watt, 120-volt, machine tool, portable, floating adjustment.
Design:	
Electrical rating:	100 watts, 120 volts.
Dimensions and configurations:	See figures 1 and 2.
Mounting base:	Material for mounting base shall be either zinc or aluminum. The base shall be not less than 3 inches wide, of a universal type designed for vertical and horizontal mounting. The base shall include two removable auxiliary "C" clamps with an opening of not less than 2 inches and a throat depth of not less than 1-3/4 inches.
Arms:	Material for arms shall be steel tubing.
Lighting head:	The lighting head shall have three aluminum shades totally shielding a 100-watt medium screw base incandescent lamp as shown on figure 2. Air venting slots shall be provided on the outer shade for cooling purposes. The shades shall be separated using nylon spacers.
Lamp:	The light shall be designed to accommodate a 100 watt, 120 volt, medium screw, inside frosted, rough service incandescent lamp in accordance with MS15541-2, FSN 6240-246-5052. Lamp shall not be furnished.
Lampholder:	The lampholder shall have a medium screw base socket with a molded phenolic shell. The socket shall have a rating of 250 volts and 250 watts. The lampholder shall be detachable and it shall be mounted in and supported by the lighting head.
Switch:	Switch shall be toggle, double pole, single throw conforming to MS16569-1, FSN 5930-719-6959.
Cable and connector plug:	Light shall be equipped with 9 feet +2 inches of cable, type SJO, AWG size 18, 3 conductor (white, black, red) in accordance with J-C-580. A three conductor plug conforming to style D of W-C-596 shall be molded to end of cable in accordance with J-C-175, type 1 assembly. All metal parts of the light shall be grounded to the red conductor and to the ground pin of the plug.
Internal wiring and grounding:	All internal wiring shall be routed within the tubular arms. Means shall be provided for positive ground connection within light for ground wire of supply cable. Positive grounding of all metal parts exposed to touch shall be maintained throughout entire light. This shall be accomplished in such a manner as to afford utmost protection to shipboard personnel. A check between outer or inner shades and ground terminal on plug shall reflect a near zero impedance.
Swivel joints:	The base, arms and lighting head shall be connected together with friction swivel joints which shall provide rotation and movement as shown on figure 2. The swivel joints shall provide sufficient friction to retain the lighting head within the tolerance specified in the vibration test.
Finish:	Finish shall be in accordance with MIL-F-16377 except that the inner shade shall be painted with a baked-on high temperature resistant white enamel.
Enclosure effectiveness:	Totally enclosed.
Weight:	5 pounds maximum.
Labels:	The following labels shall be furnished in accordance with MIL-F-16377 attached as shown on figure 2.

(a) One type I.
(b) One type IV. The information to be shown on this label shall be: "USE LAMP 6240-246-5052 ONLY."

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Quality assurance: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection: See table I
Quality conformance:
Comparison inspection: Same as the first article inspection

Inspection of product for delivery:
Examination: See MIL-F-16377
Examination for preparation of delivery: See MIL-F-16377
Quality conformance tests: See MIL-F-16377

Table I - First article and comparison inspection.

Inspection	Requirement paragraph	Inspection paragraph	Remarks
Examination	3.13	4.5	
Operation	3.13	4.8.1	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Shock	3.5.12	4.8.7	1/
Vibration	3.5.13	4.8.8	1/
Enclosure effectiveness	3.5.11	4.8.14.4	Totally enclosed
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Grounding circuit	3.6.13.3	4.8.11	

1/ The light shall be mounted as follows for this test:

(a) The lower arm shall make a 60 degree angle with the horizontal.
(b) The upper arm shall be parallel to the horizontal.
(c) The lighting head shall make a 45 degree angle with the upper arm facing downward.

Test shall be considered satisfactory if lighting head moves no more than 3 inches from placed position. Breakage of any part shall be cause for rejection.

General information: Assignment of military part number and general information is as follows:

(a) Military part number: M16377/56-206.2
(b) Federal stock number: FSN 6230-883-6209
(c) The light shown herein was previously covered by MS17268 and Drawing 9-S-4943-L.

Preparing activity:
Navy - SH
(Project 6230-N175)

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